Lecture # 7

Wednesday, 11 March 2026 1:56 pm

$$\left.\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned}\right\} \rightarrow$$

Gauss - method

Echlow

$$\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & b & c & d \\ 0 & 1 & f & g \\ 0 & 0 & 1 & h \end{array}\right) \end{array}$$

$1x_3 = h$

$\boxed{x_3 = h}$

$x_2 + f x_3 = g$

$x_2 + fh = g$

$\boxed{x_2 = g - fh}$

$1x_1 + bx_2 + cx_3 = d$

$x_1 + b(g - fh) + ch = d$

$x_1 = d - b(g - hf) - ch$

Gauss- Jordan

Row reduced Echlow

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ \left(\begin{array}{cccc} 1 & 0 & 0 & e \\ 0 & 1 & 0 & f \\ 0 & 0 & 1 & g \end{array}\right) \end{array}$$

$x_3 = g$

$x_2 = f$

$x_1 = e$

(1)

$$\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 1 \end{array}\right) \end{array}$$

Unique

(1) Echlon form

(2) $\boxed{x_3 = 1}$

$1x_2 + 2x_3 = 8$

$x_2 + 2 = 8$

$\boxed{x_2 = 6}$

$1x_1 + 2x_2 + 3x_3 = 5$

$x_1 + 12 + 3 = 5$

$x_1 = 5 - 15$

$\boxed{x_1 = -10}$

(11) Special case

$$\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right) \end{array}$$

$0x_1 + 0x_2 + 0x_3 = 0$ True

$x_2 + 2x_3 = 4$

$\boxed{x_2 = 4 - 2x_3}$

$x_1 + 2x_2 + 3x_3 = 5$

$x_1 + 2(4 - 2x_3) + 3x_3 = 5$

$x_1 + 8 - 4x_3 + 3x_3 = 5$

$x_1 + 8 - x_3 = 5$

$x_1 = x_3 - 3$

$x_1 = x_3 - 3$

$x_2 = 4 - 2x_3$

$x_3 = t$

$\{t-3, 4-2t, t\}$ for $t \in \mathbb{R}$

(2)

$$\begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & \boxed{1} & 0 & 6 \end{array}\right) \end{array}$$

$1x_2 = 6$

$\boxed{x_2 = 6}$

$1x_1 + 2x_2 + 3x_3 = 4$

$x_1 + 12 + 3x_3 = 4$

$x_1 = -3x_3 - 8$

$$\left(\begin{array}{ccc|c} 0 & \boxed{1} & 0 & 6 \end{array}\right)$$

$x_1 = -3x_3 - 8$

$x_1 = -3t - 8$

$(-3t-8,\ 6,\ t)$

(3)
$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & \boxed{6} \end{array}\right)$$

has no solution

$0x_1 + 0x_2 + 0x_3 \neq 6$ Not Possible

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned}$$

First Column

1

$0 \rightarrow R_2 \leftrightarrow R_1$

$0 \rightarrow R_3 \leftrightarrow R_1$

(1)
$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ \boxed{3} & -7 & 4 & 10 \end{array}\right)$$

R_3: $3 \quad -7 \quad 4 \quad 10$

$3R_1$: $3 \quad 3 \quad 6 \quad 24$

$0 \quad -10 \quad -2 \quad -14$

$R_2 + R_1$

$R_3 - 3R_1$
$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right)$$

$-1R_2$
$$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & \boxed{-10} & -2 & -14 \end{array}\right)$$

0

1

0

$1 \quad 0 \quad 7 \quad 17$

R_3: $0 \quad -10 \quad -2 \quad -14$

$$R_1 - R_2 \qquad R_3 + 10R_2$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right)$$

R_3: 0 $\quad -10 \quad -2 \quad -14$

$10R_2$ 0 $\quad 10 \quad -50 \quad -90$

0 $\quad 0 \quad -52 \quad -104$

$$-\frac{1}{52}R_3 \qquad \left(\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right)$$

R_1: 1 $\quad 0 \quad 7 \quad 17$

$7R_3$ 0 $\quad 0 \quad 7 \quad 14$

1 $\quad 0 \quad 0 \quad 3$

$$\begin{array}{c} \\ R_1 - 7R_3 \\ R_2 + 5R_3 \end{array} \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right) \end{array}$$

$R_2 =$ 0 $\quad 1 \quad -5 \quad -9$

$5R_3 =$ 0 $\quad 0 \quad 5 \quad 10$

0 $\quad 1 \quad 0 \quad 1$

$x_1 = 3, \quad x_2 = 1, \quad x_3 = 2$